

Design and Optimisation of a Polarizer Ring for Precision Energy Calibration of the FCC-ee Collider

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The FCC-ee collider is a proposed electron-positron collider requiring extremely precise beam energy calibration. Accurate determination of beam energy can be achieved using resonant depolarization of polarized electron bunches. While the baseline strategy is to generate polarized bunches directly in the collider, a dedicated lower-energy polarizer ring has been proposed to improve operational efficiency. This project aims to refine and optimise the design of such a polarizer ring through beam dynamics simulations in Xsuite. This will determine key parameters such as ring geometry, lattice configuration, magnet strengths, polarization time, equilibrium polarization, and chromaticity correction.

Chromaticity correction is rather part of the design than (aiming at increasing the momentum acceptance) than a key parameter

add "... and positron ..." (or remove "electron")

I. PROJECT AIMS

A. Introduction to the Project Area

The FCC-ee is a proposed 90 km circumference electron-positron collider designed to operate at multiple beam energies. Precision measurements of electroweak observables require extremely accurate beam energy determination. This can be achieved using resonant depolarization of polarized electron bunches.

In storage rings, electron beams naturally polarize through the Sokolov-Ternov effect due to synchrotron radiation. However, polarization build-up time can be long and may interfere with collider operation. A dedicated lower-energy polarizer ring has therefore been proposed to generate polarized bunches more efficiently.

Initial lattice concepts exist but require refinement in optics design, chromaticity correction, nonlinear dynamics, radiation integrals, and polarization optimization.

- Identifying limitations and potential imperfections beyond ideal simulations.

- The final outcome will be a technically documented lattice proposal with quantified performance characteristics.

C. Motivation and Beneficiaries

The project is of interest to the FCC-ee collaboration, with the potential for the design to become a baseline concept for further accelerator development. A dedicated polarizer ring could increase efficiency of collider operation by separating polarization generation from physics data-taking.

"could" => "would"?

The primary beneficiary is the FCC-ee design study within CERN's BE-ABP-INC group.

Remove this part (the FCC study is above and relies on the work of many sections and groups). You could add here or somewhere else a comment that FCC is studied to be proposed as the next CERN flagship project.

B. Description of Aims and Objectives

The overall aim is to design and characterize a viable polarizer ring lattice using numerical simulations. Objectives include:

- Constructing and testing lattice configurations in Xsuite.
- Determining magnet placement and strengths (dipoles, quadrupoles, sextupoles).
- Evaluating and correcting chromaticity.
- Studying nonlinear beam dynamics and dynamic aperture.
- Estimating polarization build-up time and equilibrium polarization.

Bullet "Evaluation of basic properties for a realistic ring with imperfections as alignment errors and unwanted magnet multipolar components and their correction" or something similar instead of listing imperfections as separate bullet point later? Then, the next two points would become sub-bullet points,

II. PROJECT PLAN

A. Lattice Design and Simulation

The work will be carried out primarily using Xsuite. A baseline lattice will be constructed and analysed to determine Twiss parameters, tunes, and chromaticity.

B. Chromaticity and Nonlinear Optimisation

Sextupoles will be implemented to correct chromaticity. Tracking simulations will be used to study nonlinear beam dynamics and assess dynamic aperture.

C. Radiation and Polarization Studies

Synchrotron radiation effects will be included to estimate polarization build-up time and equilibrium polarization. Where feasible, spin tracking tools in Xsuite will be used to evaluate polarization performance.

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Misalignments and their correction should be implemented quite soon, just after first dynamic aperture and momentum acceptance studies. Do you need help for re-arranging and writing?

NB: Without closed orbit perturbations, polarization build-up studies do not make much sense.

D. Imperfections and Sensitivity Studies

If time permits, potential physical imperfections may be explored.

III. PROJECT DEPENDENCIES AND TRAINING NEEDS

The successful execution of this project depends on access to Xsuite and sufficient computational resources

I would re-order a bit the topics under III: First general accelerator physics and beam dynamics, then the particular accelerator physics domains quoted and, finally, the simulation tools (may-be to some extent matter of taste?).

for detailed lattice optimisation and spin tracking simulations. It also requires advanced understanding of synchrotron radiation theory, beam dynamics, polarization mechanisms, and accelerator physics to interpret and validate simulation results.

IV. RISK REGISTER

Potential risks include insufficient dynamic aperture, long polarization build-up time, or excessive chromaticity. These risks will be mitigated through lattice optimisation and parameter studies.

A risk is that, contrary to results from initial preliminary studies, sufficient polarization cannot be maintained up to the collider (e.g., injection into collider not yet studied).